THE CATEGORY OF FORMAL ALGEBRAIC SPACES HAS FIBRE PRODUCTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof span cites two same-chapter lemmas, both delivered as navigation stubs. No bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The fibre product of two formal algebraic spaces over a sheaf whose diagonal is representable by algebraic spaces. Same chapter, Lemma lemma-fibre-products-general, invoked at proof lines 3526-3527. Provides the existence statement once its hypothesis on the base is verified.
- The diagonal of a formal algebraic space is representable by algebraic spaces. Same chapter, Lemma lemma-diagonal-formal-algebraic-space, invoked at proof lines 3528-3529. Verifies the hypothesis of the general lemma for the terminal object, so that the special case applies.

Source reference. Original lemma 0AIP: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2667–2667. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AJ2: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3488–3488. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. The category of formal algebraic spaces over S has fibre products.*

Proof. Special case of Lemma 0AJ2 because formal algebraic spaces have representable diagonals, see Lemma 0AIP. □

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